



Automatic ECG signal quality assessment using convolutional neural networks and derivative ECG signal for false alarm reduction in wearable vital signs monitoring devices[☆]

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ABSTRACT

The electrocardiogram (ECG) signals are often analyzed to detect cardiovascular diseases and monitor vital signs. However, analysis of noisy ECG signals leads to misdiagnosis of diseases and generates false alarms. To prevent false alarms, we present a derivative ECG (dECG) signal-based lightweight convolutional neural network (CNN) for automatic ECG signal quality assessment (ECG-SQA). The proposed CNN detects clean (“acceptable”) and noisy (“unacceptable”) ECG signals which ensures only clean ECG signals are analyzed for disease detection and monitoring vital signs with reduced false alarms in health monitoring devices. Here, we evaluated the performance, total parameters, testing time for ECG-SQA, and model size of 60 dECG-based CNNs to determine the optimal ECG-SQA method. The performance of the dECG-based CNNs are analyzed with three activation functions, five kernel sizes, different numbers of convolutional layers, and dense layers. The CNNs are trained using ECG signals from one channel and fifteen channels of standard ECG databases. On a standard unseen ECG database, the proposed CNN model has achieved accuracy, sensitivity, and specificity of 97.59%, 98.78%, and 89.23%, respectively. The optimal CNN (model size: 2,989 kB) implemented on the Raspberry Pi computing platform has testing time of 130.44 ± 46.24 ms for quality assessment of 5 s ECG signal which confirms the real-time feasibility of the proposed method. The dECG-based ECG-SQA method is essential during continuous monitoring of vital signs and diagnosis of cardiovascular disease to reduce false alarms and improve reliability of wearable devices having limited computing capacity and onboard memory.

1. Introduction

Electrocardiogram (ECG) signals represent the electrical activity of the heart. The ECG signals are analyzed to diagnose cardiovascular diseases and monitor vital signs. In hospitals, ECG signals are recorded in a controlled environment to prevent contamination by various noises, such as baseline wander, muscle artifacts, electrode movement, and powerline interference. The development of wearable technology has enabled individuals to record ECG signals without expert intervention in an active environment. Wearable devices have sensors that capture ECG signals from the body surface. However, the sensors are susceptible to noises or artifacts, which alter the morphological features of the ECG signals, such as the P wave, QRS complex, and ST segment. The analysis of noisy ECG signals with distorted morphology results in false alarms and misdiagnosis of diseases [1–5]. Moving average filtering, frequency selective filtering, adaptive filtering, and discrete wavelet

transform fail to remove noises that have morphological features similar to ECG without distorting the shape, amplitude, and duration of the actual signal [6]. Therefore a lightweight, robust, and real-time implementable ECG signal quality assessment (ECG-SQA) method is required to detect clean and noisy ECG signals in wearable devices with limited computing resources. The ECG-SQA method reduces false diagnosis of disease and false alarms by detecting noisy ECG signals before parameter extraction during long-term, continuous monitoring of ECG signals in a physically active environment.

1.1. Related works and limitations

In this section, we discussed the existing machine learning-based ECG-SQA methods. The performances of these methods were evaluated using the ECG signals from the PhysioNet/Computing in Cardiology

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Challenge 2011 (PCCC2011) database. Kuvzilek et al. [7] presented a rule-based support vector machine (SVM) for ECG quality assessment method. The testing accuracy (AC) of the method was 83.6%. In [8], Kalkstein et al. implemented a quasi-linear combination of k-nearest neighbors (KNN) and random forest classifier for ECG-SQA. The method reported AC of 92.95% and false positivity rate of 5.85%. In [9], Zhou et al. presented an one-dimensional (1D) convolutional neural network (CNN) to classify ECG signals as acceptable and unacceptable, achieving AC, sensitivity (SE), and specificity (SP) of 94.3%, 91.3%, and 95.5%, respectively, when tested single-lead ECG signals of duration 10 s. Tan et al. [10] proposed a neural network-based ECGQAnet for ECG-SQA. The method had AC, SE, and SP of 94.55%, 85.68%, and 97.02%, respectively. Jin et al. [11] presented a deep learning-based dual-attentional convolutional long short-term memory (DAC-LSTM) neural network for evaluating ECG signal quality. The reported AC, SE, and SP of the method were 94%, 97.59%, and 76.47%, respectively. Liu et al. [12] implemented CNN with Stockwell transform-based ECG-SQA. The method had mean AC of 93.09% and SE of 97.67%. Mondal et al. in [13] presented a 1D CNN-based ECG-SQA method with ECG signals as input to the CNN model. The reported AC, SE, and SP of the method were 90.66%, 92.88%, and 75%, respectively. In [14], Zhu et al. proposed an AdaBoost-based method for detecting ECG signal quality. The method reported test AC of 95.8%. Several methods used the PhysioNet/Computing in Cardiology Challenge 2017 (PCCC2017) database [15] for training and testing. In [16], Zhang et al. reported a comparison among the random forest, kernel SVM (KSVM), least square SVM (LS-SVM), and multi-surface proximal SVM-based ECG-SQA method. The LS-SVM had the highest AC of 92.2% compared to the random forest, KSVM, and multi-surface proximal SVM-based ECG-SQA methods. Huerta et al. [17] implemented 2D CNN (AlexNet) to detect the quality of ECG signals. The reported AC, SE, and SP of the ECG quality detection method were 91.2%, 100%, and 90.3%, respectively, and the test time was 17.29 s. In another study Huerta et al. [18] compared pre-trained 2D CNNs, namely AlexNet, Visual Geometry Group 16 (VGG16), and GoogLeNet-based ECG quality assessment methods. The GoogLeNet had the highest AC of $90.75 \pm 0.75\%$, SE of $88.8 \pm 2.77\%$, and SP of $92.7 \pm 1.72\%$. In [19], Huerta et al. trained and tested AlexNet to detect the quality of the ECG signals. The AlexNet-based method had AC, SE, and SP of 91.4%, 90.4%, and 93.2%, respectively. A comparison among the pre-trained 2D CNN models AlexNet, VGG16, GoogLeNet, ResNet18, and InceptionV3-based ECG-SQA were presented by Huerta et al. in [20]. The study showed that AlexNet had the least testing time of 12.76 ± 0.21 s compared to VGG16, GoogLeNet, ResNet18, and InceptionV3-based methods. The InceptionV3-based ECG-SQA method had the highest AC of $91.25 \pm 0.88\%$, SE of $89 \pm 2.24\%$, and SP of $93.5 \pm 0.79\%$. In [21], Seeuws et al. presented an autoencoder-based ECG-SQA method. The reported SE and SP of the method were 96% and 70%, respectively. Herraiz et al. [22] proposed a 2D CNN-based ECG-SQA method studied on three databases, namely PCCC2017, Public Telehealth (THDB), and a proprietary database. The reported AC of the method was $88.95 \pm 1.03\%$, $91.42 \pm 1.08\%$, and $94.79 \pm 1.41\%$ when tested with PCCC2017, THDB, and proprietary databases, respectively. In [4], Mondal et al. implemented 1D CNN-based method for detecting the quality of ECG signals. The 1D CNN had testing AC of 100% when trained and tested with ECG from the Massachusetts Institute of Technology (MIT)-Boston's Beth Israel Hospital (BIH) arrhythmia (MIT-BIHA) database.

In most existing studies, the same database was considered for training and testing the machine learning models in ECG-SQA methods. As the signals have the similar diagnostic and morphological features in the training and testing dataset, the SQA methods achieved high detection accuracy. Also, most existing studies did not compute testing time for ECG-SQA and did not consider the real-time feasibility of the ECG-SQA methods in their analysis.

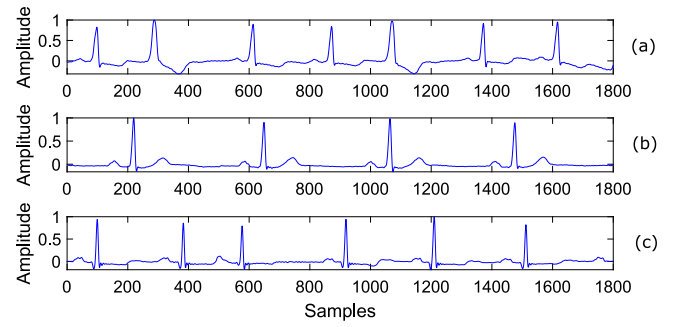


Fig. 1. Clean ECG signals with arrhythmia from the MIT-BIHA database record (a) 221, (b) 202, and (c) 205.

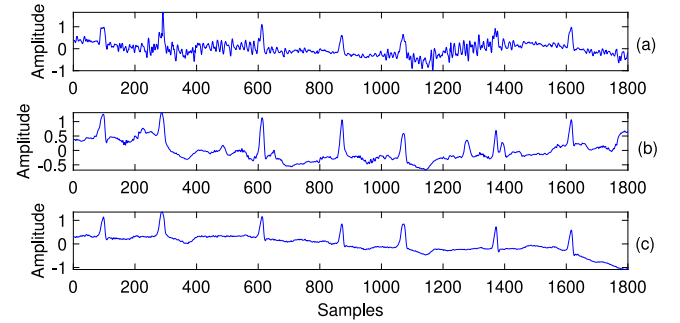


Fig. 2. ECG signals taken from the MIT-BIHA database contaminated with (a) muscle artifacts, (b) electrode movement, and (c) baseline wander.

1.2. Major contributions and organisation of this paper

In this paper, our objective is to develop an automatic ECG-SQA method to detect clean (“acceptable”) and noisy (“unacceptable”) ECG signals, shown in Figs. 1 and 2, respectively. In ECG-based wearable health monitoring devices, the feature analysis of noisy ECG signals generates false alarms due to misdiagnosis of disease. The major contributions of this paper are summarized as follows:

- Presented the least computationally complex first-order derivative of ECG (dECG) signal-based lightweight CNN for detecting the quality of ECG signals as clean (“acceptable”) and noisy (“unacceptable”).
- Evaluated the performance of 60 dECG-based CNN models with different hyperparameters, including activation function, convolutional layer (CL), dense layer (DL), and kernel size to identify the optimal CNN model with a high SQA performance, smaller model size, and lower testing time.
- Tested the robustness of the dECG-based CNNs with St. Petersburg Institute of Cardiological Technics (INCART) 12-lead arrhythmia database [23] and PCCC2011 database [24], which are unseen to the trained dECG-based CNN models.
- Implemented the optimal CNN-based ECG-SQA on the Raspberry Pi (R-Pi) (1.5 GHz Cortex-A72 quad-core CPU and 8 GB RAM) to test the feasibility of the method on a low computational-resource platform.

The rest of the paper is organized as follows. Section 2 has materials and methods. The results and discussion of the proposed method, which includes real-time implementation and computational complexity analysis of the method are presented in Section 3. Section 4 provides the conclusion.

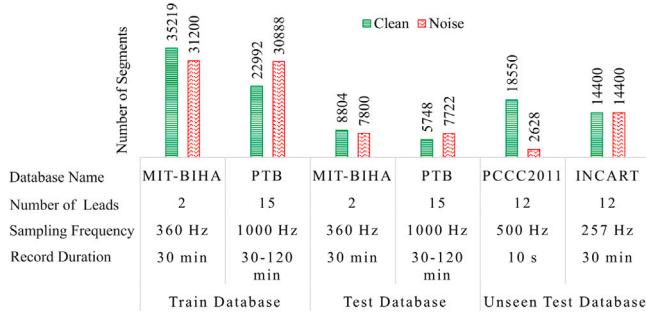


Fig. 3. Summary of ECG databases used to train, test, and unseen test the 1D CNN models in this paper. Here, min denotes minute.

2. Materials and methods

The automatic ECG-SQA is required to reduce false alarms and improve the reliability of wearable devices during continuous monitoring of vital signs and detection of cardiac diseases [7,8,13]. We propose a dECG-based 1D CNN for the quality assessment of ECG signals in resource-constrained battery-powered wearable devices. This section includes the train and test database description and the proposed methodology.

2.1. Database description

In this paper, the CNNs are trained with ECG signals from the MIT-BIHA database [25] and the Physikalisch-Technische Bundesanstalt diagnostic ECG (PTB) database [26]. The MIT-BIHA is a 2-channel, 48-record database having different ECG morphologies, including normal sinus rhythm, atrial fibrillation, ventricular trigeminy, ventricular tachycardia, sinus arrhythmia, and also noisy signals. The ECG signals are collected from 47 subjects. The records are of 30 min and have a sampling frequency of 360 Hz. The PTB database is a 15-channel database, having 549 records collected from 290 subjects. The diagnostic classes of the subjects majorly include myocardial infarction, cardiomyopathy, bundle branch block, and dysrhythmia. The records are of 30–120 min and have a sampling frequency of 1000 Hz. The unseen test databases INCART and PCC2011 are used to evaluate the performance of the ECG-SQA method. The INCART 12-lead arrhythmia database contains 75 records, collected from 32 subjects having mostly ventricular ectopic beats. The records are of 30 min each and sampled at 257 Hz. The PCC2011 validation set consists of 1000 12-lead ECG records, each of 10 s, and sampled at 500 Hz. Fig. 3 summarizes the databases considered to train, test, and unseen test the proposed dECG-based CNNs for ECG-SQA.

The block diagram in Fig. 4 illustrates the proposed methodology for ECG-SQA using the dECG-based CNN. The block diagram can be broadly divided into three sections: preprocessing of ECG signals, creation of clean and noisy ECG database, and training and testing of the CNNs for quality assessment of ECG signals.

2.2. Preprocessing

The ECG signals taken from the four standard databases have different recording setups, including the number of leads, sampling frequency, and recording duration, as shown in Fig. 3. Therefore, preprocessing the signals is essential before using them in training and testing of the CNN models. This subsection presents the steps that are followed before database creation.

Resampling and segmentation: The databases considered in this paper have different sampling frequencies due to different recording setups. The ECG signals from the PTB database, PCC2011, and INCART databases are resampled to 360 Hz from their original sampling rates of 1000 Hz, 500 Hz, and 257 Hz, respectively. All training and test signals are split into non-overlapping 5 s signals. The non-overlapping segmentation resulted in 17,280 and 16,470 ECG signals of 5 s each from the MIT-BIHA and PTB databases, respectively. The 30 min long muscle artifact, baseline wander, and electrode movement records from the MIT-BIH noise stress test database (MIT-BIH NSTDB) [27] is split into 5 s noise segments to contaminate the ECG signals with different noises.

Filtering: During recording, ECG signals often get corrupted by various noises and artifacts, such as baseline wander, electrode movement, muscle artifacts, and powerline interference. The ECG signals are distorted by the denoising methods implemented to remove artifacts having morphological features similar to ECG signals [3]. We have applied a second-order high-pass Chebyshev filter with a cutoff frequency of 0.8 Hz and a fourth-order low-pass Chebyshev filter with a cutoff frequency of 40 Hz to remove baseline wander and high-frequency noises, respectively [13]. The filtering is performed in both forward and reverse directions to avoid phase distortion of the signals. In this paper, ECG signals containing artifacts in the frequency range of 0.8–40 Hz are detected as noisy ECG signals using the ECG-SQA method.

Normalization: The ECG signals are amplitude normalized to avoid the influence of signal amplitude variations caused by the different recording setups of the databases used in training and testing the CNNs [28]. The amplitude normalization of ECG signals and noise segments between -1 to 1 is computed as

$$y[n] = \frac{x[n] - \mu_x}{\max(|x[n] - \mu_x|)} \quad (1)$$

where $x[n]$ is the input signal ($n = 1, 2, \dots, 1800$), μ_x is the mean of all the sample values of the input signal, and “max” computes the maximum of the absolute value of the signal after subtracting the mean.

2.3. Database creation and labeling

The normalized ECG signals from the MIT-BIHA and PTB databases are visually analyzed to remove signals with abrupt changes and severe noises. As a result, 11,006 and 14,370 clean 5 s ECG signals are obtained from the MIT-BIHA and the PTB databases, respectively. Since there is an imbalance between the clean (25 376) and noisy (8374) ECG signals and CNN model training requires a large training set, hence we created the noisy ECG database and augmented the clean ECG signals.

The noisy ECG database is created using 500 and 495 clean 5 s ECG signals from the MIT-BIHA and the PTB databases, respectively. These 995 ECG signals have different morphological features of the QRS complex, P waves, and ST segments from different leads (modified limb lead II and 15 leads). Fig. 1 presents the clean ECG signals from the MIT-BIHA database. The powerline interferences are generated of unit amplitude, sampled at 360 Hz, and of frequencies 50 Hz and 60 Hz using MATLAB. Baseline wander, muscle artifacts, and electrode movement noise segments from the noise records of the MIT-BIH NSTDB and powerline interference are attenuated to the factor of 0.3, 0.6, and 0.9 before contaminating the 995 clean ECG signals according to the process expressed as

$$NECG = NFECG + a \times X_n \quad (2)$$

where NFECG is the noise-free (clean) ECG signal (5 s), a is the attenuation factor, X_n is the artifact of 5 s and sampled at 360 Hz. NECG is the corrupted (noisy) 5 s ECG signal. A total of 77,610 noisy ECG signals are generated by adding noise to the ECG signals. Fig. 2 shows the ECG signals contaminated with muscle artifacts, electrode movement, and

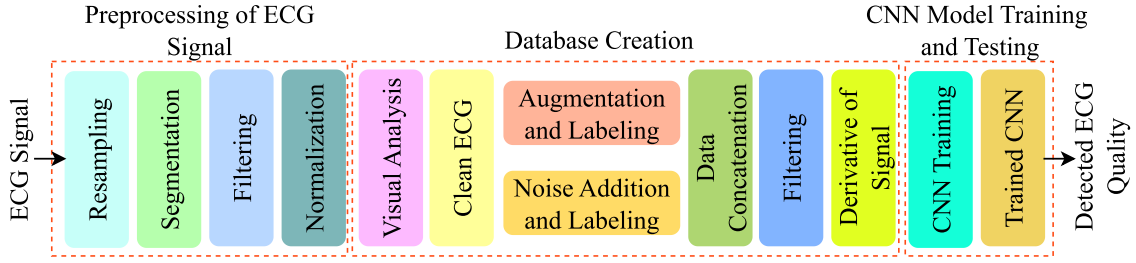


Fig. 4. ECG signal quality assessment based on dECG and 1D CNN.

Table 1

Training, test, and unseen test data for dECG-based CNN.

Dataset information	Specifications
Format	.mat
Number of channels	1, 12, and 15
ECG sampling frequency	360 Hz
ECG segment duration (Training and testing)	5 s and 5 s
Training data (Clean: Noisy)	58 211:62 088
Test data (Clean: Noisy)	14 552:15 522
Unseen test data (PCCC2011 and INCART) (Clean: Noisy)	18 550:26 28 and 14 400:14 400

baseline wander. The noisy ECGs are filtered using high-pass and low-pass Chebyshev filters with cutoff frequencies of 0.8 Hz and 40 Hz, respectively. The noisy ECG signals are annotated as 1 for training the CNN models, shown in Fig. 5. A random signal of 1800 sample size is generated using the rand(1800,1) function in MATLAB. The signal is amplitude normalized between -1 and 1 , attenuated to 0.001 , 0.002 , 0.003 , and 0.004 and added to all 11,006 and 7185 clean ECG signals taken from the MIT-BIH arrhythmia and PTB databases, respectively. The augmentation process is expressed as

$$\text{NFECG}_b = \text{NFECG} + b \times R_{1800} \quad (3)$$

where NFECG is the clean ECG signal, R_{1800} is the normalized random signal generated in MATLAB, b is the attenuation factor, and NFECG_b is the augmented clean ECG signal. The augmentation process created a total of 72,763 clean ECG signals. Clean ECG signals are annotated as 0 for training the CNN models. The number of clean and noisy train, test, and unseen test ECG signals are shown in Table 1.

2.4. Derivative of ECG (dECG)

Before training and testing the CNNs, the first-order derivative of the signals $y[n]$ is computed as [29]

$$d[n] = y[n+1] - y[n] \quad (4)$$

The derivative of the ECG signals alters the amplitude of the components of the signal based on frequency. The high-frequency components (noise) present in the ECG signals are enhanced with respect to the low-frequency components [30]. The amplitude of the high-frequency low amplitude noises present in the ECG signals are enhanced by the first-order derivative. Fig. 6 shows the derivative of the ECG signals with muscle artifacts, electrode movement, and baseline wander. The dECG of the signal is given as input to CNN models [31], which detect the quality of the ECG signal.

2.5. CNN model

In this paper, 60 dECG-based CNN models are developed by tuning the hyperparameters to automatically detect the quality of ECG signals on a resource-constrained platform. The CNNs are trained with six

Table 2

Hyper-parameters of dECG-based 1D CNN models.

Parameters	Specifications
Input size	1×1800
1D convolutional layers	2 3 4 5
Max-pooling layers	2 3 4 5
Dropout layers	2 3 4 5
Dense layers	3 3 and 4 4 and 5 5
Activation functions	Leaky ReLU, ReLU, and ELU
Kernel sizes	3×1 , 4×1 , 5×1 , 6×1 , and 7×1
Batch size, learning rate	10, 0.001
Optimizer	Adam
Loss function, epochs	Binary cross-entropy, 100

different combinations of the CLs and DLs (2 CLs and 3 DLs, 3 CLs and 3 DLs, 3 CLs and 4 DLs, 4 CLs and 4 DLs, 4 CLs and 5 DLs, and 5 CLs and 5 DLs). The models are evaluated with five kernel sizes (3×1 , 4×1 , 5×1 , 6×1 , and 7×1), and three different activation functions, namely rectified linear unit (ReLU), leaky ReLU, and exponential linear unit (ELU). The number of kernels in the 1st, 2nd, 3rd, 4th, and 5th CLs are set to 8, 16, 32, 64, and 128, respectively. Each CL is followed by a max-pooling layer of size 3×1 with a stride of 2 and a dropout layer with a rate of 0.3. A flatten layer (FL) is there in between CLs and DLs. The neurons in the five consecutive DLs are in decreasing order 64, 32, 16, 8, and 1 (output DL). The CNN serves as a binary classifier with a single neuron in the output layer, activated by the sigmoid activation function, to classify clean and noisy ECG signals. The total number of parameters and the performance of each CNN depends on the number of CLs and DLs, as well as the kernel size [32]. The activation function also influences the performance of CNN models. In wearable health monitoring devices, the ECG-SQA method with high detection accuracy must be lightweight, which is rarely addressed in existing literature.

Table 2 presents the hyperparameters of the CNN models. A batch size of 10 is considered during training, with the learning rate set to 0.001. The adaptive moment estimation (Adam) [4] optimizer and binary cross-entropy loss function are used in training the CNNs for 100 epochs. The training is conducted on the Google Colab platform using a graphics processing unit (GPU). Fig. 5 illustrates a CNN model with 4 CLs and 4 DLs. The consecutive CL contain 8, 16, 32, and 64 kernels. Each CL is followed by a max-pooling layer of size 3×1 and a dropout layer with a rate of 0.3. The flatten layer converts the rectangular feature matrix into a column matrix [33–35]. This CNN model includes 32, 16, 8, and 1 neurons in the four consecutive DLs, with the output of the flatten layer serving as input to the DLs.

3. Results and discussion

The dECG-based CNNs are evaluated with the test database (20% of the processed ECG data) generated from the MIT-BIHA and the PTB databases and two unseen test databases, PCCC2011 and INCART. The performance of the CNN models are evaluated using the benchmark metrics, SE, SP, and AC, which are computed as [3,36]

$$\text{SE} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\% \quad (5)$$

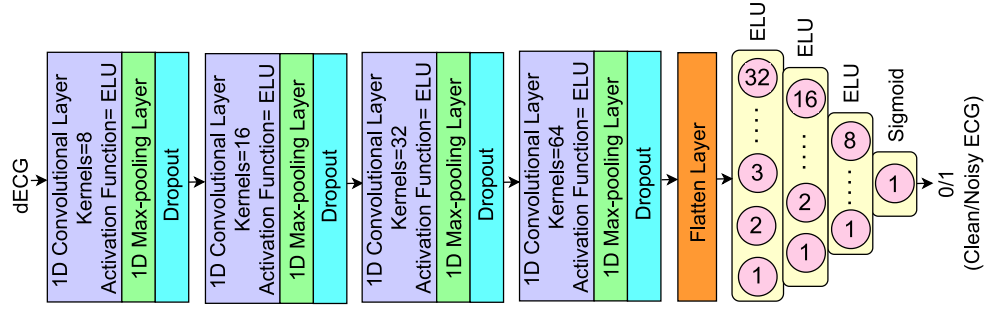


Fig. 5. The CNN with 4 convolutional layers, ELU activation function, kernel size 5×1 , 4 max-pooling layers of size 3 and stride 2, dropout of 0.3, and 4 dense layers.

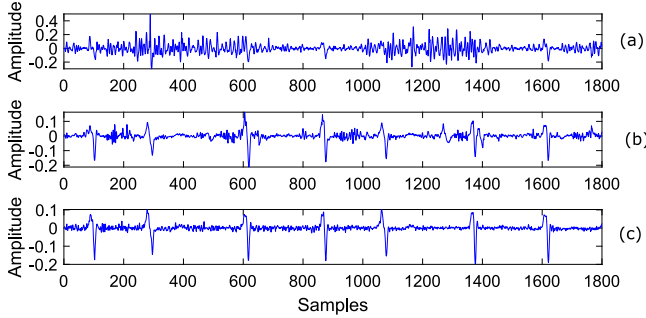


Fig. 6. Derivative of filtered ECG signals from Fig. 2; (a) muscle artifacts, (b) electrode movement, and (c) baseline wander.

$$SP = \frac{TN}{TN+FP} \times 100\% \quad (6)$$

$$AC = \frac{TP+TN}{TP+TN+FP+FN} \times 100\% \quad (7)$$

where true positive (TP) is the correctly detected clean ECG as clean, false negative (FN) is the incorrectly detected clean ECG as noisy, true negative (TN) is the correctly detected noisy ECG as noisy, and false positive (FP) is the incorrectly detected noisy ECG as clean. The CNNs are analyzed based on the performance with different activation functions, kernel sizes, and combinations of CLs and DLs. To find the best models for each activation function, the CNNs are compared in terms of performance metrics and model size. The CNNs are implemented on R-Pi to evaluate the feasibility of the proposed method in real-time. We computed the testing time required for quality assessment of 5 s ECG signal on a resource-constrained platform to find the optimal CNN model. Finally, the performance of dECG-based optimal lightweight CNN for ECG-SQA is compared with existing machine learning-based ECG quality assessment methods.

3.1. Performance evaluation

The dECG-based CNNs are evaluated for ECG-SQA using three different activation functions, five different kernel sizes, four different numbers of CLs, and three different DLs.

3.1.1. Performance of CNN models with ReLU, leaky ReLU, and ELU activation functions

Table 3 summarizes ECG-SQA performance, total parameters, and model size of the dECG-based CNNs with ReLU activation function and kernel sizes 3×1 , 4×1 , and 5×1 . The CNN with kernel size 5×1 , 4 CLs, and 4 DLs achieved the highest performance when tested with the PCCC2011 database. Conversely, the CNN with kernel size 4×1 , 5 CLs, and 5 DLs exhibited the best performance when tested with the INCART database. A performance comparison between the two models shows that, although the 5×1 CNN has 8.82% lower SP on the INCART

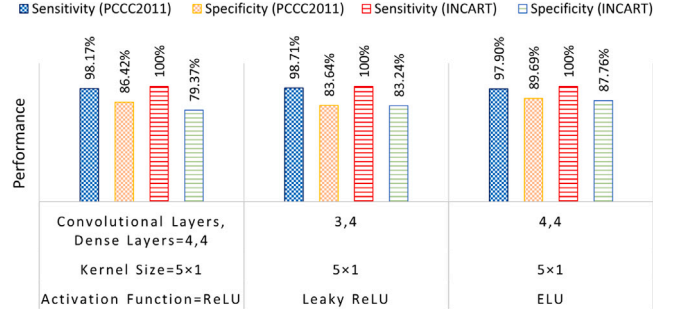


Fig. 7. Best Performing dECG-based SQA method for ReLU, leaky ReLU, and ELU activation with optimal kernel size, convolutional and dense layers.

database, its SE is 30.31% higher when tested with the PCCC2011 database. Furthermore, the model size of the kernel 5×1 CNN with 4 CLs and 4 DLs decreased by 51.63% compared to the kernel 4×1 CNN with 5 CLs, and 5 DLs.

Table 4 presents the performance, total parameters, and model size of dECG-based CNNs with the leaky ReLU activation function and kernel sizes 3×1 , 4×1 , and 5×1 . The CNN with kernel size 5×1 , 3 CLs, and 4 DLs achieved the best performance when tested with the PCCC2011 database. However, the CNN with kernel size 5×1 , 4 CLs, and 4 DLs performed the best when tested with the INCART database. The performance comparison shows that the SP of the CNN with 3 CLs and 4 DLs improved by 4.81% than the CNN with 4 CLs and 4 DLs when tested with the INCART database. Moreover, the 3 CLs and 4 DLs CNN is with 11 328 less parameters than the CNN with 4 CLs and 4 DLs.

Table 5 summarizes the performance, total parameters, and model size of the dECG-based CNNs with ELU activation function and kernel sizes 3×1 , 4×1 , and 5×1 . The CNN with kernel size 5×1 , 4 CLs, and 4 DLs performed the best when tested with the PCCC2011 database. However, the CNN with kernel size 4×1 , 4 CLs, and 5 DLs achieved the best performance when tested on the INCART database. Test results with PCCC2011 show that the kernel 5×1 CNN has 25.36% higher SE than the kernel 4×1 CNN. Moreover, the model size of the CNN with kernel size 5×1 , 4 CLs, and 4 DLs is 47.81% lower than the model size of CNN with kernel size 4×1 , 4 CLs, and 5 DLs.

Fig. 7 summarizes the performance of the best CNN models with each of the ReLU, leaky ReLU, and ELU activation functions. The CNN with ELU activation has 3.27% and 6.05% higher SP than the best CNNs with ReLU and leaky ReLU activation functions, respectively, when tested on the PCCC2011 database. On the INCART database, the CNN with ELU activation function showed SP improvements of 8.39% and 4.52% compared to the CNNs with ReLU and leaky ReLU activation functions, respectively.

Fig. 8 presents the SP of 45 dECG-based CNNs tested with 20% of the processed ECG signals from MIT-BIHA and PTB datasets. The CNN with the ELU activation function and kernel size 3×1 achieved SP of

Table 3
SQA performance of dECG-based CNNs with ReLU activation function and different kernel sizes.

Activation function	Kernel size	CNN (CL, DL)	Unseen database						20% of processed database			Total parameters	Model size (kB)
			PCCC2011 database			INCART database			MIT-BIHA and PTB database				
			AC (%)	SE (%)	SP (%)	AC (%)	SE (%)	SP (%)	AC (%)	SE (%)	SP (%)		
ReLU	3 × 1	2,3	86.53	92.36	45.40	66.47	98.74	34.19	98.54	100	97.18	1,15,793	1408
		3,3	92.11	98.15	49.51	67.59	100	35.18	93.67	100	87.75	1,17,361	1434
		3,4	88.77	96.63	33.26	57.76	99.94	15.58	81.40	100	64.03	2,33,105	2800
		4,4	89.71	97.71	33.22	58.13	99.96	16.31	87.94	100	76.68	2,40,337	2894
		4,5	89.39	99.02	21.35	51.66	99.99	3.33	81.51	100	64.23	4,73,873	5636
		5,5	86.56	93.74	35.88	50.44	97.89	3.00	88.00	99.21	77.53	5,02,673	5987
	4 × 1	2,3	92.34	96.16	65.33	81.06	99.96	62.16	99.84	100	99.70	1,15,929	1410
		3,3	93.68	98.60	58.90	70.14	99.97	40.32	96.20	100	92.66	1,18,009	1442
		3,4	93.75	97.65	66.25	74.67	99.85	49.48	99.68	100	99.38	2,33,753	2808
		4,4	91.73	99.40	37.60	57.17	99.95	14.38	84.99	100	70.97	2,43,033	2926
		4,5	90.96	97.58	44.18	55.48	99.44	11.52	88.80	100	78.34	4,76,569	5668
		5,5	70.40	67.86	88.32	92.15	96.12	88.19	99.49	98.97	99.97	5,13,561	6115
	5 × 1	2,3	95.47	98.16	76.48	89.78	99.99	79.56	100	100	100	1,16,065	1412
		3,3	96.77	98.81	82.34	89.34	100	78.67	99.96	100	99.92	1,18,657	1450
		3,4	95.34	97.26	81.85	87.81	99.97	75.65	100	100	99.99	2,34,401	2816
		4,4	96.71	98.17	86.42	89.68	100	79.37	99.96	100	99.93	2,45,729	2958
		4,5	96.63	98.53	83.22	86.99	100	73.97	99.87	100	99.75	4,79,265	5700

Table 4
SQA performance of dECG-based CNNs with leaky ReLU activation function and different kernel sizes.

Activation function	Kernel size	CNN (CL, DL)	Unseen database						20% of processed database			Total parameters	Model size (kB)
			PCCC2011 database			INCART database			MIT-BIHA and PTB database				
			AC (%)	SE (%)	SP (%)	AC (%)	SE (%)	SP (%)	AC (%)	SE (%)	SP (%)		
Leaky ReLU	3 × 1	2,3	89.74	93.50	63.20	78.73	99.79	57.67	99.94	100	99.88	1,15,793	1408
		3,3	93.42	97.87	61.99	77.60	100	55.21	99.42	100	98.89	1,17,361	1434
		3,4	90.69	98.22	37.52	63.91	99.96	27.86	94.06	100	88.51	2,33,105	2800
		4,4	92.10	99.12	42.54	63.63	100	27.25	94.36	100	89.09	2,40,337	2894
		4,5	91.16	96.68	52.13	71.49	99.97	43.01	96.72	100	93.65	4,73,873	5636
		5,5	93.94	97.23	70.74	80.75	99.99	61.50	99.78	100	99.57	5,02,673	5987
	4 × 1	2,3	93.24	96.89	67.47	82.41	99.92	64.91	99.95	100	99.90	1,15,929	1410
		3,3	95.61	97.91	79.34	87.39	100	74.78	100	100	100	1,18,009	1442
		3,4	95.90	99.22	72.49	81.94	100	63.88	99.61	100	99.25	2,33,753	2808
		4,4	94.64	95.96	85.27	91.20	99.99	82.42	99.99	99.97	100	2,43,033	2926
		4,5	93.77	97.62	66.59	77.80	99.99	55.60	99.43	100	98.91	4,76,569	5668
		5,5	95.44	98.89	71.04	80.14	100	60.28	99.09	100	98.24	5,13,561	6114
	5 × 1	2,3	95.12	98.64	70.32	82.50	99.99	65.02	99.97	100	99.94	1,16,065	1412
		3,3	97.41	99.40	83.41	90.65	100	81.29	99.99	100	99.98	1,18,657	1450
		3,4	96.84	98.71	83.64	91.62	100	83.24	100	100	99.99	2,34,401	2816
		4,4	97.06	98.80	84.82	89.22	100	78.43	99.99	100	99.99	2,45,729	2958
		4,5	96.16	98.23	81.51	87.55	100	75.10	99.97	100	99.95	4,79,265	5700

Table 5
SQA performance of dECG-based CNNs with ELU activation function and different kernel sizes.

Activation function	Kernel size	CNN (CL, DL)	Unseen database						20% of processed database			Total parameters	Model size (kB)
			PCCC2011 database			INCART database			MIT-BIHA and PTB database				
			AC (%)	SE (%)	SP (%)	AC (%)	SE (%)	SP (%)	AC (%)	SE (%)	SP (%)		
ELU	3 × 1	2,3	85.61	89.16	60.58	74.50	99.15	49.85	100	100	99.99	1,15,793	1408
		3,3	94.14	97.68	69.14	84.23	100	68.45	99.99	100	99.99	1,17,361	1434
		3,4	93.88	98.85	58.75	77.75	100	55.50	99.05	100	99.05	2,33,105	2796
		4,4	96.06	97.88	83.26	89.47	100	78.94	100	100	100	2,40,337	2894
		4,5	95.12	98.64	70.28	81.72	100	63.44	88.29	100	77.35	4,73,873	5636
	4 × 1	2,3	91.13	94.92	64.42	78.20	99.90	56.51	99.82	100	99.65	1,15,929	1410
		3,3	96.28	99.04	76.83	88.89	100	77.77	99.95	99.97	99.94	1,18,009	1442
		3,4	95.63	97.82	80.21	90.68	99.97	81.40	99.97	99.97	99.97	2,33,753	2804
		4,4	96.64	97.83	88.28	93.20	99.97	86.43	99.96	99.97	99.94	2,43,033	2926
		4,5	75.14	72.54	93.49	97.95	98.32	97.58	97.46	94.73	100	4,76,569	5668
		5,5	65.10	60.94	94.41	95.98	92.64	99.31	94.14	87.87	100	5,13,561	6110
	5 × 1	2,3	91.16	94.27	69.18	80.08	99.64	60.51	99.93	99.97	99.89	1,16,065	1412
		3,3	96.78	99.53	77.36	88.07	100	76.13	99.94	100	99.88	1,18,657	1450
		3,4	97.19	99.53	80.67	90.06	100	80.13	100	100	99.99	2,34,401	2812
		4,4	96.88	97.90	89.69	93.88	100	87.76	99.92	99.97	99.86	2,45,729	2958
		4,5	95.43	99.44	67.12	78.47	100	56.94	99.92	99.97	99.86	4,79,265	5700

Table 6

The SQA performance, testing time in PC-CPU, total parameters, and model size of dECG-based CNN models with ELU activation and kernel sizes 6×1 and 7×1 .

Activation function	Kernel size	CNN (CL, DL)	Unseen database								Total parameters	Model size (kB)
			PCCC2011 database				INCART database					
			AC (%)	SE (%)	SP (%)	Testing time (ms)	AC (%)	SE (%)	SP (%)	Testing time (ms)		
ELU	6 × 1	2,3	90.45	92.89	73.21	66.54 ± 12.83	84.39	99.21	69.57	62.68 ± 11.26	1,16,201	1413
		3,3	96.62	98.91	80.44	67.67 ± 15.63	90.44	99.99	80.90	63.66 ± 14.17	1,19,305	1457
		3,4	95.91	98.17	79.95	67.39 ± 15.43	89.85	99.97	79.73	64.11 ± 12.80	2,35,049	2819
		4,4	97.59	98.78	89.23	66.80 ± 16.71	94.33	100	88.66	64.57 ± 14.91	2,48,425	2989
		4,5	96.91	98.56	85.27	70.09 ± 23.44	93.42	99.99	86.84	66.31 ± 21.48	4,81,961	5731
ELU	7 × 1	2,3	91.00	93.19	75.49	68.35 ± 14.44	85.02	99.39	70.66	63.03 ± 12.26	1,16,337	1415
		3,3	96.09	97.41	86.76	68.80 ± 14.06	93.90	99.92	87.87	63.61 ± 13.65	1,19,953	1,465
		3,4	96.10	98.16	81.58	66.93 ± 14.56	91.13	99.93	82.33	63.47 ± 12.64	2,35,697	2827
		4,4	94.84	95.62	89.35	69.75 ± 22.17	94.61	99.76	89.45	64.81 ± 27.82	2,51,121	3021
		4,5	96.94	98.19	88.09	68.49 ± 16.03	93.69	99.99	87.38	66.83 ± 15.81	4,84,657	5763

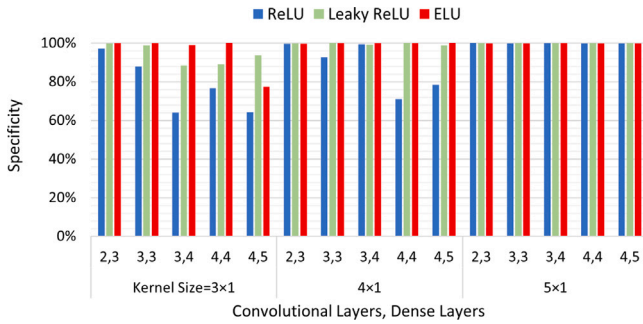


Fig. 8. Specificity of dECG-based CNNs; for kernel size $3 \times 1, 4 \times 1$ and 5×1 ; ReLU, leaky ReLU, and ELU activation function, different number of CLs and DLs ((2,3), (3,3), (3,4), (4,4), and (4,5)) tested with 20% of processed signal from MIT-BIH arrhythmia and PTB database.

100%. Besides, the SP of the CNNs with ReLU and leaky ReLU activation functions improved with the increase in kernel size. All the CNNs, regardless of the activation functions, CLs, and DLs combinations, achieved 100% SP with kernel size 5×1 .

3.1.2. Performance of CNNs with ELU activation function and kernel size 6×1 and 7×1

The comparison of ECG-SQA performance indicates that the dECG-based CNN with ELU activation outperformed the CNNs with ReLU and leaky ReLU activation functions. Further analysis is conducted to evaluate the performance of CNNs with ELU activation and kernel sizes 6×1 and 7×1 , shown in Table 6. Evaluation results show that the ECG-SQA performance of the CNN with kernel size 6×1 , 4 CLs, and 4 DLs achieved 3.16% higher SE than CNN with kernel size 7×1 , 4 CLs, and 4 DLs. Moreover, the SE and SP of the CNN with kernel size 6×1 improved by 0.88% and 0.9%, respectively, compared to the best CNN with kernel size 5×1 and ELU activation, when tested on PCC2011 and INCART databases. The visual representation of the CNN features of the unseen PCC2011 and INCART databases are shown in Fig. 9(a) and (b), respectively, obtained by using the 3-dimensional t-distributed stochastic neighbor embedding (t-SNE) feature visualization [37,38]. The visualizations reveal that the dECG-based CNN can effectively distinguish between the clean (indicated in black) and noisy (indicated in red) ECG signals.

The quality assessment and intermediate steps of two long-duration ECG signals of 60 s, taken from the MIT-BIHA database (record numbers 208 and 223), are presented in Figs. 10 and 11, respectively. The predicted and actual quality of the signal is represented by a solid line (black) and a dashed line (red), respectively. The clean and noisy ECG signals are indicated by 0 and 1, respectively. In Fig. 10(c), a comparison of the predicted and actual quality of the ECG signals shows that all seven noisy ECG signals are correctly detected as noisy signals.

Out of the five clean ECG signals, four are correctly detected as clean. In Fig. 11(c), a comparison of the actual and predicted quality of the signal shows that all four noisy ECG signals are correctly detected as noisy signals, and out of eight clean signals, seven are detected as clean. The ECG-SQA method reduces false alarms by correctly detecting and discarding noisy ECG signals from vital signs monitoring. All the noisy ECG signals in Figs. 10 and 11 are correctly detected as noisy by the optimal dECG-based CNN, reducing the false alarm rate to zero.

3.2. Real-time implementation and computational complexity analysis

Limited battery resource, computing capacity, and onboard memory are constraints of wearable continuous health monitoring devices [39]. The real-time feasibility of the dECG-based CNN is evaluated by computing the performance and testing time for ECG-SQA on R-Pi as a resource-constrained computing platform [32], shown in Fig. 12. Table 7 presents the SE, SP, and testing time of the CNNs with optimal kernel size, the optimal number of CLs, and DLs with three different activation functions. The ELU-based CNN achieved the best ECG-SQA performance compared to the ReLU-based and leaky ReLU-based CNNs. The model size of the ELU-based CNN and its testing time for quality assessment of a 5 s ECG signal on R-Pi is 2989 kB and 130.44 ± 46.24 ms, respectively. The computational complexity of the proposed ECG-SQA method is demonstrated by comparing the performance and testing time of the method implemented on PC-CPU (Processor: Intel(R) Core(TM) i5-3470 CPU 3.20 GHz, RAM: 16 GB) as resource-efficient and R-Pi (Processor: 64-bit quad-core ARM Cortex-A72, RAM: 8 GB) as resource-constrained platforms. This paper shows that performance in resource-efficient and resource-constrained computing platforms remained unchanged, shown in Table 7. However, the testing time increased by 63.64 ms when implementing the optimal dECG-based CNN on the resource-constrained platform compared to that on the resource-efficient platform.

Fig. 13 presents a 60 s ECG signal recorded in the lab setup (RMS Vega 204 cardiac stress test system, frequency response: 0.005 Hz to 100 Hz) using three electrodes (placed at left wrist, right wrist, and left lower abdomen). The predicted quality of the signal by the dECG-based CNN is indicated by green line, and the actual quality is indicated by red dashed line. The method correctly detected six clean and four noisy ECG signals out of the seven clean and five noisy ECG signals.

3.3. Comparison with existing methods

Table 8 presents the performance and testing time of the proposed and existing machine learning-based ECG-SQA methods, tested with the PCC2011 database. Most of the existing ECG-SQA methods have training data from the PCC2011 database. Therefore, ECG signals with the same morphological or diagnostic feature in both training and testing phase of the machine-learning-based models resulted in the high performance of the existing SQA methods [40–43]. This paper aims to develop a robust ECG-SQA method to detect clean and noisy

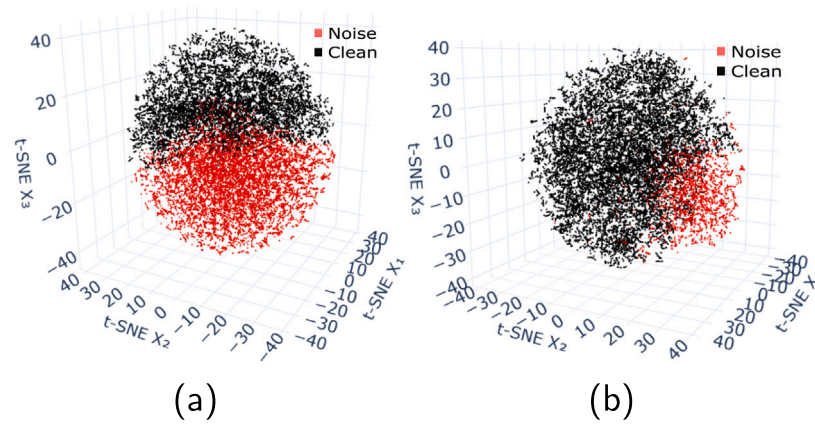


Fig. 9. The 3D t-distributed stochastic neighbor embedding (t-SNE) feature visualization results of the dECG-based optimal CNN tested with (a) PCC2011 and (b) INCART database to demonstrate effectiveness of the proposed dECG and CNN features.

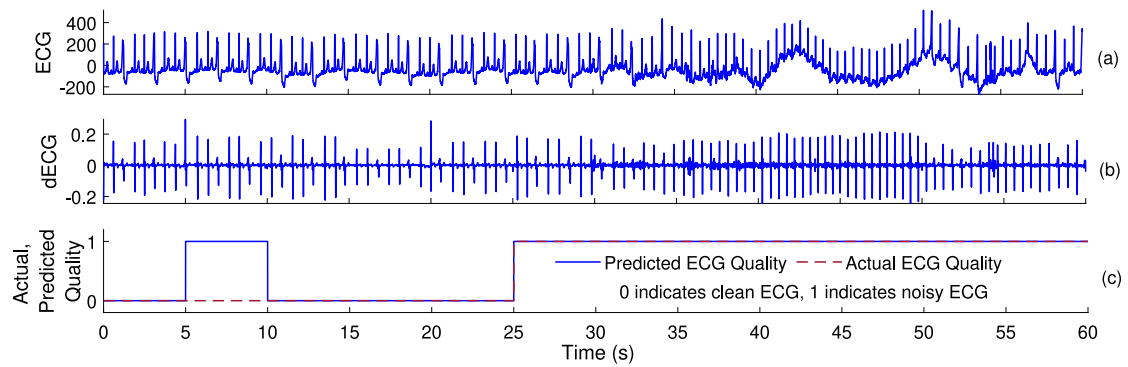


Fig. 10. (a) 60 s of continuous ECG signal taken from MIT-BIHA database record 208, (b) first-order derivative of filtered output of ECG in (a), and (c) actual quality and predicted ECG quality by the proposed model for 5 s ECG segment. Label 1 and label 0 indicate noisy and clean ECG signals, respectively.

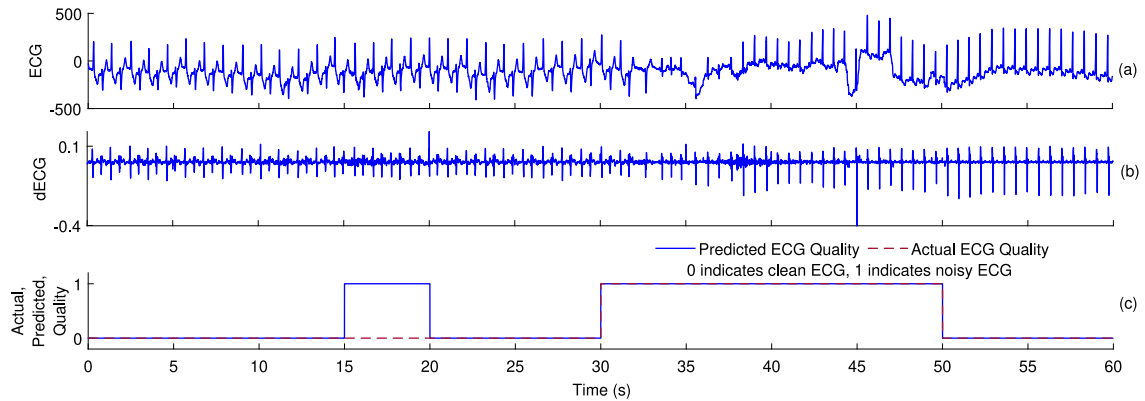


Fig. 11. (a) 60 s ECG signal taken from MIT-BIHA database record 223, (b) first-order derivative of filtered output of ECG in (a), and (c) actual quality and the predicted ECG quality by the proposed model for 5 s ECG segment. Label 1 and label 0 indicate noisy and clean ECG signals, respectively.

Table 7

Best Performing dECG-based SQA models for ReLU, leaky ReLU, and ELU activation and their testing time in PC-CPU and R-Pi, total parameters, and model size.

Activation function	Kernel size	CNN (CL, DL)	Unseen database								TotalParameters	Model size (kB)				
			PCCC2011 database						INCART database							
			SE		SP		Testing time (ms)		SE				SP		Testing time (ms)	
			(%)	(%)	(%)	(%)	PC-CPU	R-Pi	(%)	(%)			PC-CPU	R-Pi		
ReLU	5 × 1	4,4	98.17	86.42	69.19 ± 16.88	130.49 ± 41.25	100	79.37	62.98 ± 14.77	123.14 ± 40.35	2,45,729	2958				
Leaky ReLU	5 × 1	3,4	98.71	83.64	67.14 ± 14.87	134.74 ± 46.81	100	83.24	67.82 ± 14.25	122.72 ± 39.13	2,34,401	2816				
ELU	6 × 1	4,4	98.78	89.23	66.80 ± 16.71	130.44 ± 46.24	100	88.66	64.57 ± 14.91	122.19 ± 40.07	2,48,425	2989				

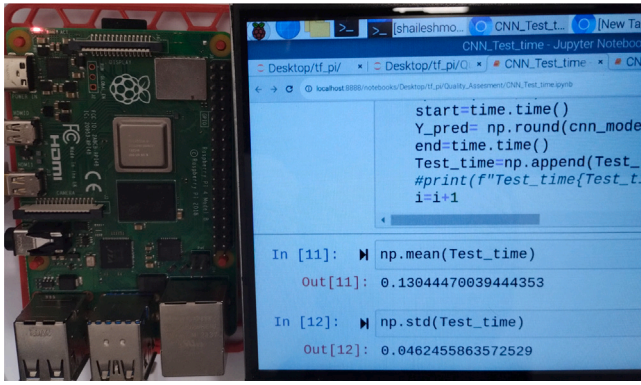


Fig. 12. Implementation of dECG-based optimal CNN for ECG-SQA on R-Pi computing platform.

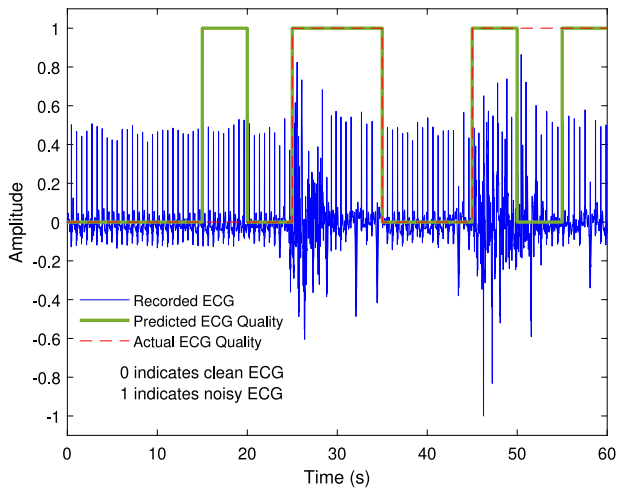


Fig. 13. Actual quality and predicted quality of ECG signal recorded in lab setup. Label 1 and label 0 indicate noisy and clean ECG signals, respectively.

Table 8

Comparison of ECG-SQA methods tested on PCC2011 database.

Reference	Method	Number of layers	AC (%)	SE (%)	SP (%)	Test time CPU (ms)
[7]	SVM	–	83.6	–	–	–
[8]	KNN and random forest	–	92.95	–	–	–
[10]	1D CNN	–	93.50	88.46	95.27	78 (Android emulator (PIXEL 30))
[11]	DAC-LSTM	10	94.0	97.59	76.47	3.43 (NVIDIA GTX 1080 GPU)
[12]	2D CNN	4	93.1	97.7	77.3	NR
[13]	1D CNN	9	90.66	92.88	75	65.6 ± 22 (Intel i5)
[14]	AdaBoost	–	95.8	98.6	87.2	NR
[16]	LS-SVM	–	92.20	–	–	NR
Proposed	1D CNN	8	97.59	98.78	89.23	66.80 ± 16.71 (Intel i5), 130.44 ± 46.24 (R-Pi)

Note: Intel i5: Intel(R) i5 Core (TM) i5-3470.

ECG signals across different recording conditions and morphological features. In this paper, the PCC2011 database is unseen to the dECG-based CNN models. The AC of the proposed method improved by 1.79% when compared with the AdaBoost-based ECG-SQA method [14]. The SP of the dECG-based CNN improved by 12.76%, 11.93%, 14.23%, and 2.03% compared to the DAC-LSTM-based [11], 2D-CNN-based [12], 1D CNN-based [13], and AdaBoost-based [14] ECG-SQA methods, respectively. The testing time for ECG-SQA of a 5 s signal is 130.44 ± 46.24 ms by the dECG-based CNN implemented on R-Pi as a resource-constrained computing platform. However, the DAC-LSTM based ECG-SQA method in [11] reported a testing time of 3.43 ms on a resource-efficient NVIDIA GTX 1080 GPU platform.

4. Conclusion

This paper presents a lightweight and real-time implementable 1D CNN-based ECG-SQA method to reduce false alarms during continuous and long-term monitoring of cardiac health and vital signs in ECG-based wearable devices. Evaluation results demonstrate that a lightweight and high-performing 1D CNN model can be achieved with an optimal selection of hyperparameters, namely the number of CLs and DLs, kernel size, and activation function. The CNN model trained with the derivative of ECG signals exhibits the best ECG-SQA performance with ELU activation function, kernel size 6×1 , 4 CLs, and 4 DLs. The optimal CNN with a model size of 2989 kB, achieved the SP or false alarm reduction rate of 89.23% and 88.66% when tested with two standard unseen ECG databases. Moreover, the paper demonstrates that the optimal 1D CNN can be implemented on R-Pi (Processor: 64-bit quad-core ARM Cortex-A72, RAM: 8 GB), maintaining same ECG-SQA performance compared to the same model implemented on PC-CPU (Processor: Intel(R) Core(TM) i5-3470 CPU 3.20 GHz, RAM: 16 GB). The real-time feasibility of the best 1D CNN model is demonstrated by its implementation on R-Pi, achieving a testing time of 130.44 ± 46.24 ms for quality assessment of 5 s ECG signal. The dECG-based ECG-SQA method significantly reduces false alarms in wearable health monitoring devices under noisy recording conditions.

CRedit authorship contribution statement

Achinta Mondal: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **M. Sabarimalai Manikandan:** Writing – review & editing, Supervision, Resources, Funding acquisition, Formal analysis, Conceptualization. **Ram Bilas Pachori:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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